

Design and Analysis of an 8-bit CMOS Arithmetic Logic Unit Using Traditional Multiplexer Architecture and an Innovative DEMUX-Based Power-Gated Architecture

Final VLSI Design Report

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Chapter 1

Introduction

1.1 Background

Arithmetic Logic Units (ALUs) are among the most important computational components in digital systems and processors. They perform arithmetic and logical operations required for data processing and computation.

Conventional ALUs generally compute all operations simultaneously and utilize multiplexers for output selection. Although widely used, this design introduces unnecessary switching activity and increased power consumption.

As modern VLSI systems increasingly prioritize low-power operation, reducing switching activity has become a major design objective.

1.2 Motivation

Traditional ALUs continuously power all arithmetic and logical blocks regardless of whether the operation is selected.

This causes:

- Unnecessary dynamic switching activity
- Increased power consumption
- Larger transistor count
- Increased propagation delay due to multiplexers

This project introduces a DEMUX-based power-gated ALU architecture in which only the selected operation block receives VDD supply.

The proposed architecture aims to:

- Reduce switching activity

- Reduce power consumption
- Eliminate multiplexer overhead
- Improve Power Delay Product (PDP)
- Reduce propagation delay

1.3 Objectives

The objectives of this project are:

1. Design transistor-level CMOS arithmetic and logical blocks
2. Implement an 8-bit traditional multiplexer-based ALU
3. Implement a DEMUX-based power-gated ALU
4. Compare both architectures
5. Analyze power, delay, and PDP

Chapter 2

Theory and Literature Review

2.1 Arithmetic Logic Units

An Arithmetic Logic Unit is a combinational digital circuit capable of performing arithmetic and logical operations.

The major blocks of a conventional ALU include:

- Arithmetic Unit
- Logical Unit
- Multiplexer
- Control Circuitry

2.2 CMOS Logic

CMOS logic is widely used because of:

- Low static power dissipation
- High noise immunity
- Reliable switching
- Scalability

A CMOS gate consists of:

- PMOS pull-up network
- NMOS pull-down network

2.3 Dynamic Power Consumption

Dynamic power in CMOS circuits is given by:

$$P = \alpha C_L V_{DD}^2 f$$

where:

- α = switching activity
- C_L = load capacitance
- V_{DD} = supply voltage
- f = switching frequency

Reducing switching activity directly reduces power consumption.

2.4 Power Gating

Power gating disconnects inactive blocks from the supply voltage.

Advantages include:

- Lower leakage power
- Reduced dynamic switching
- Improved energy efficiency

Chapter 3

ALU Operations and Functional Requirements

3.1 Supported Operations

The ALU supports eight operations selected using a 3-bit control signal.

Table 3.1: ALU Operations

OP[2:0]	Operation	Type
000	Addition	Arithmetic
001	Subtraction	Arithmetic
010	Increment	Arithmetic
011	Decrement	Arithmetic
100	NOT	Logical
101	NOR	Logical
110	NAND	Logical
111	XOR	Logical

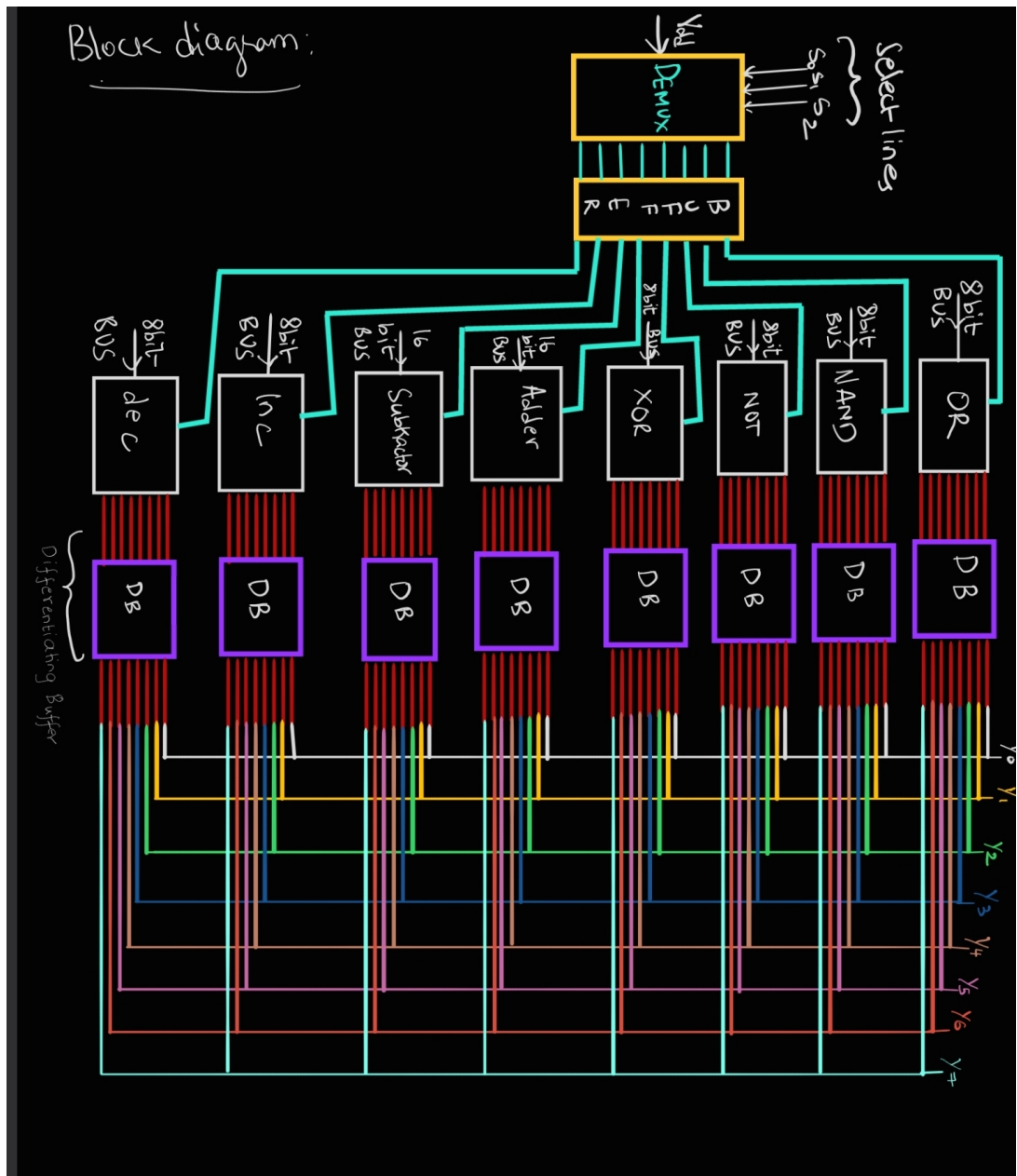


Figure 3.1: Novel DEMUX-Based Power-Gated ALU Architecture

Chapter 4

Design of Arithmetic Blocks

4.1 Full Adder

The full adder is the primary arithmetic building block used in the ALU.

$$Sum = A \oplus B \oplus C_{in}$$

$$Carry = AB + BC_{in} + AC_{in}$$

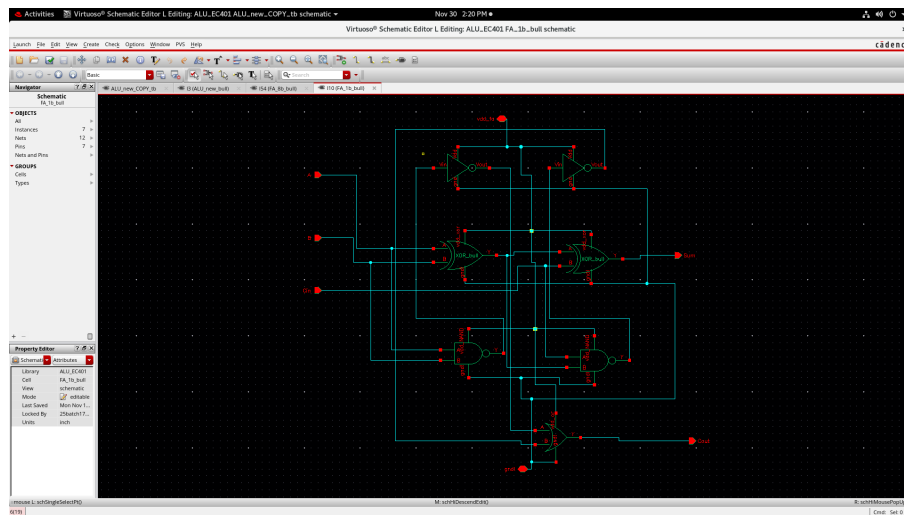


Figure 4.1: 1-Bit Full Adder

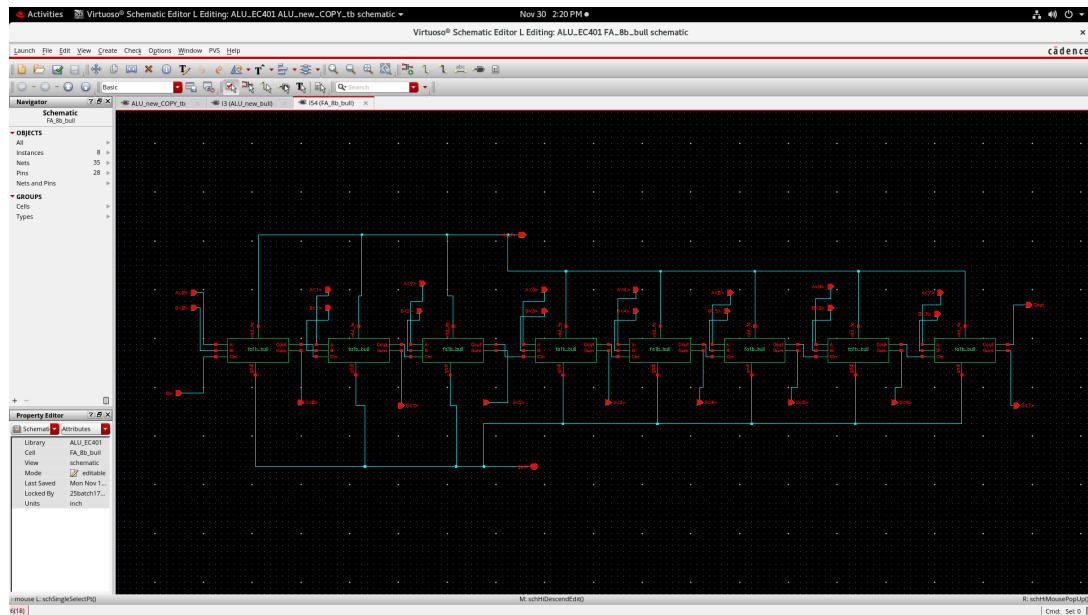


Figure 4.2: 8-Bit Ripple Carry Adder

4.2 Full Subtractor

The subtractor performs binary subtraction considering borrow propagation.

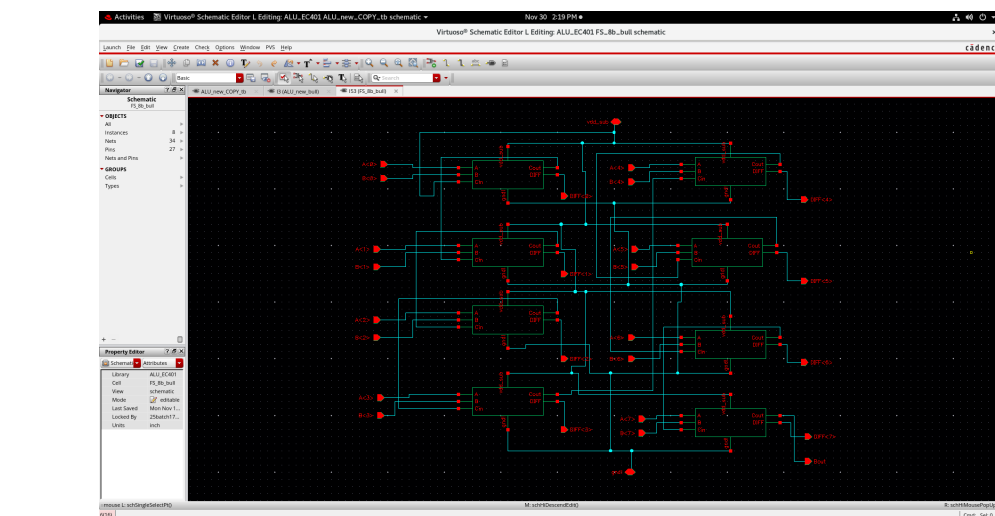


Figure 4.3: Full Subtractor

4.3 Incrementer

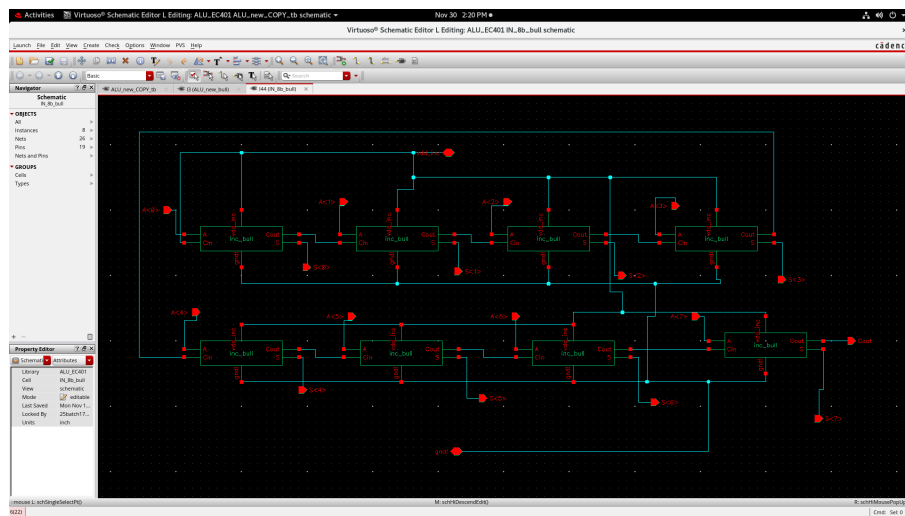


Figure 4.4: Incrementer Circuit

4.4 Decrementer

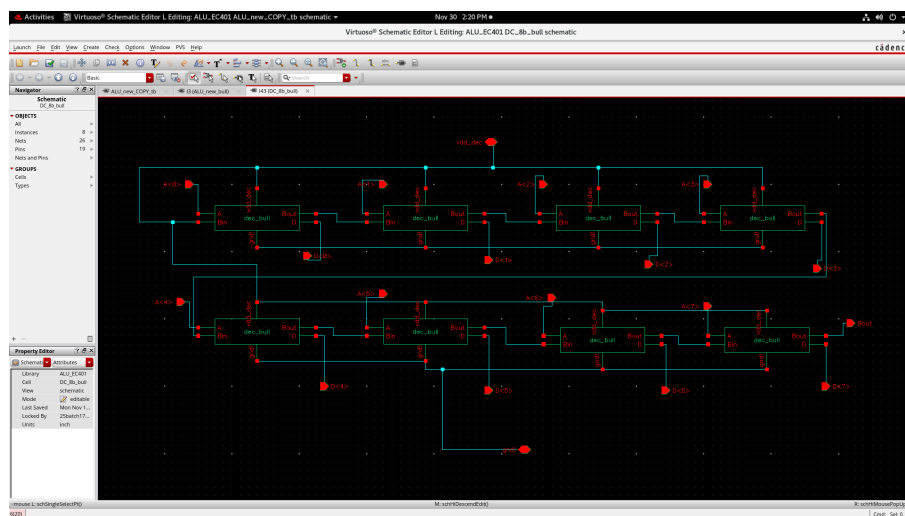


Figure 4.5: Decrementer Circuit

Chapter 5

Design of Logical Blocks

5.1 NOT Gate

The CMOS inverter produces the complement of the input signal.

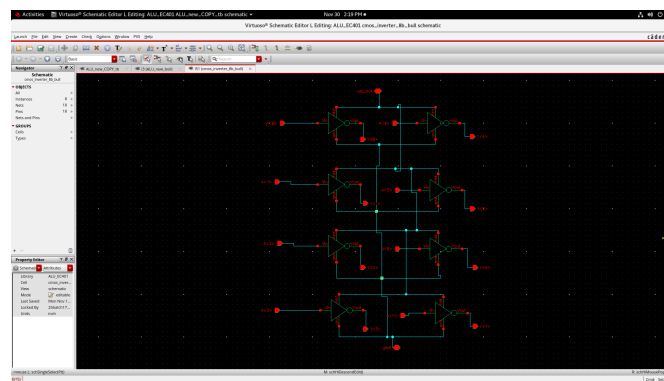


Figure 5.1: CMOS NOT Gate

5.2 NOR Gate

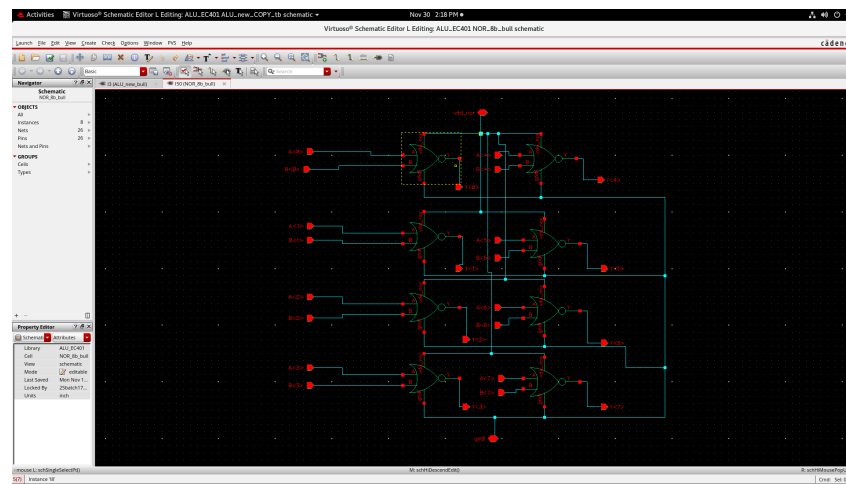


Figure 5.2: NOR Gate

5.3 NAND Gate

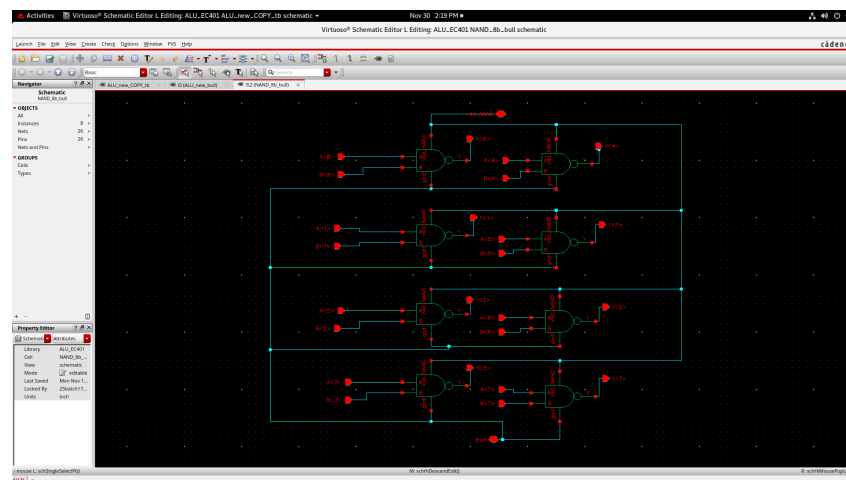


Figure 5.3: NAND Gate

5.4 XOR Gate

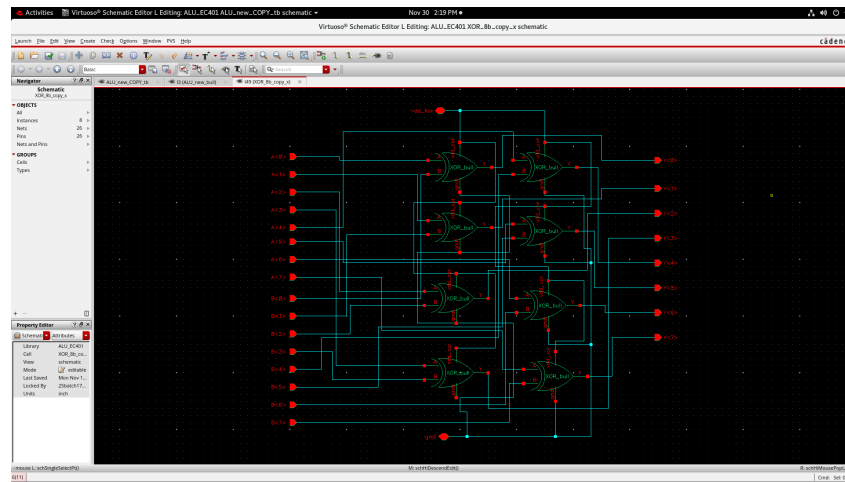


Figure 5.4: XOR Gate

Chapter 6

Traditional Multiplexer-Based ALU

6.1 Traditional Architecture

The conventional ALU computes all arithmetic and logical operations simultaneously and selects the required output using multiplexers.

Every block receives:

- Input signals
- Supply voltage
- Switching transitions

This results in unnecessary switching activity.

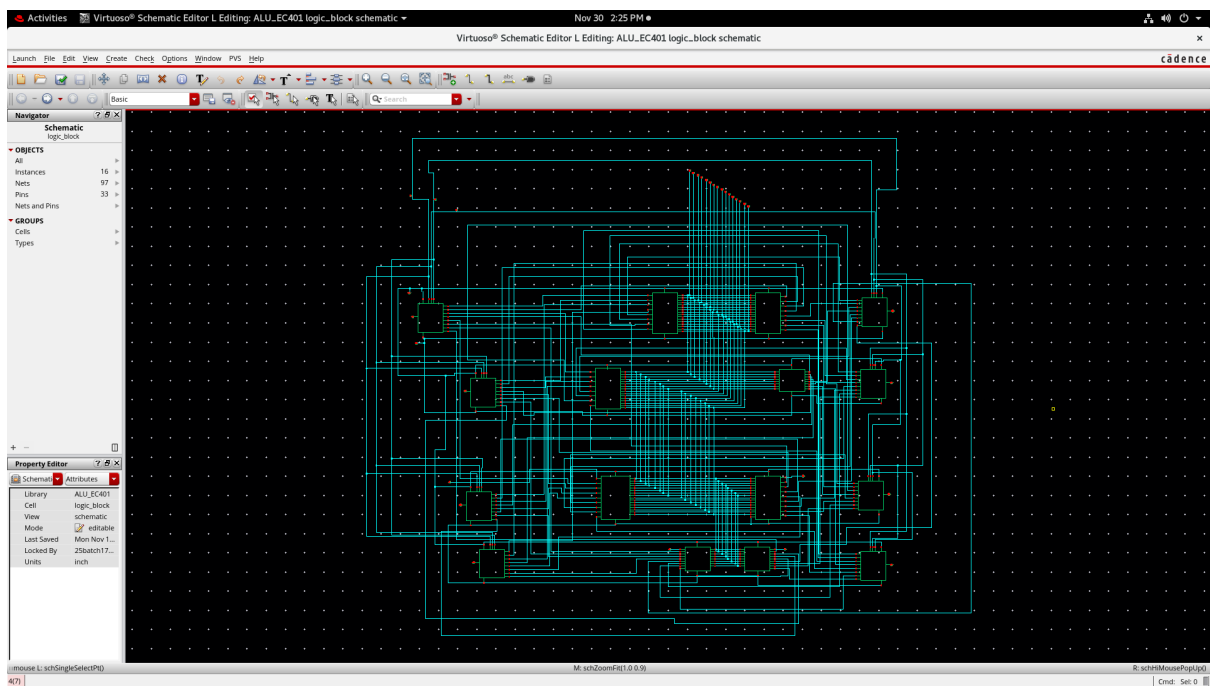


Figure 6.1: Traditional Multiplexer-Based ALU

6.2 Limitations

Major limitations include:

- High dynamic power
- Increased switching activity
- Large transistor count
- Additional propagation delay

Chapter 7

Proposed DEMUX-Based Power-Gated Architecture

7.1 Core Idea

The proposed architecture distributes VDD selectively using a DEMUX instead of selecting outputs through multiplexers.

Only the selected block receives supply voltage.

7.2 DEMUX Power Gating

A 1×8 DEMUX distributes VDD to the required arithmetic or logical block.

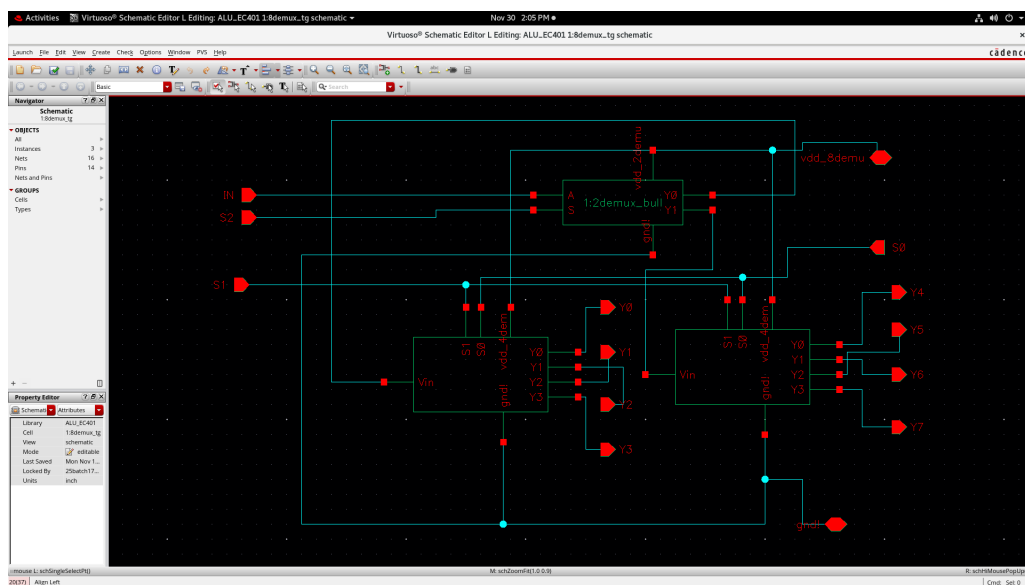


Figure 7.1: DEMUX-Based Power Distribution Circuit

7.3 Voltage Separator Circuit

The voltage separator prevents reverse voltage propagation into inactive blocks.

It also enables output merging between active blocks.

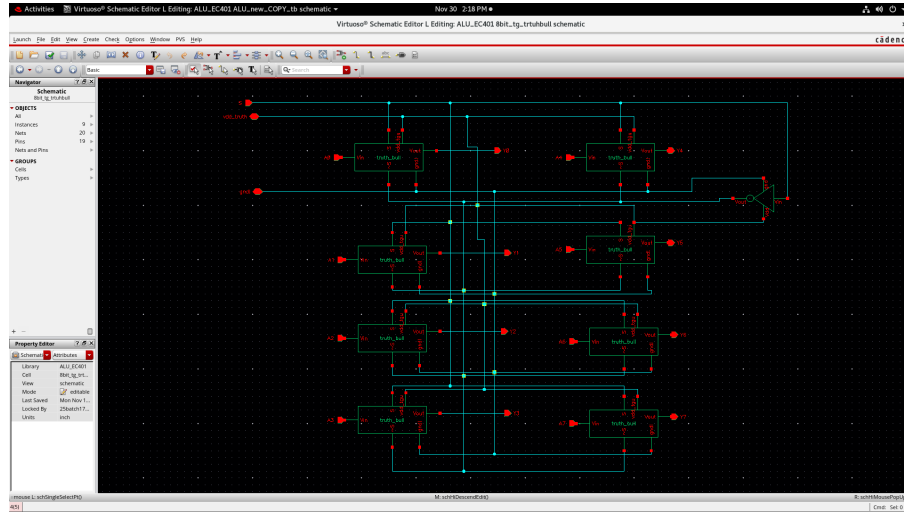


Figure 7.2: Voltage Separator Circuit

7.4 Complete DEMUX-Based ALU

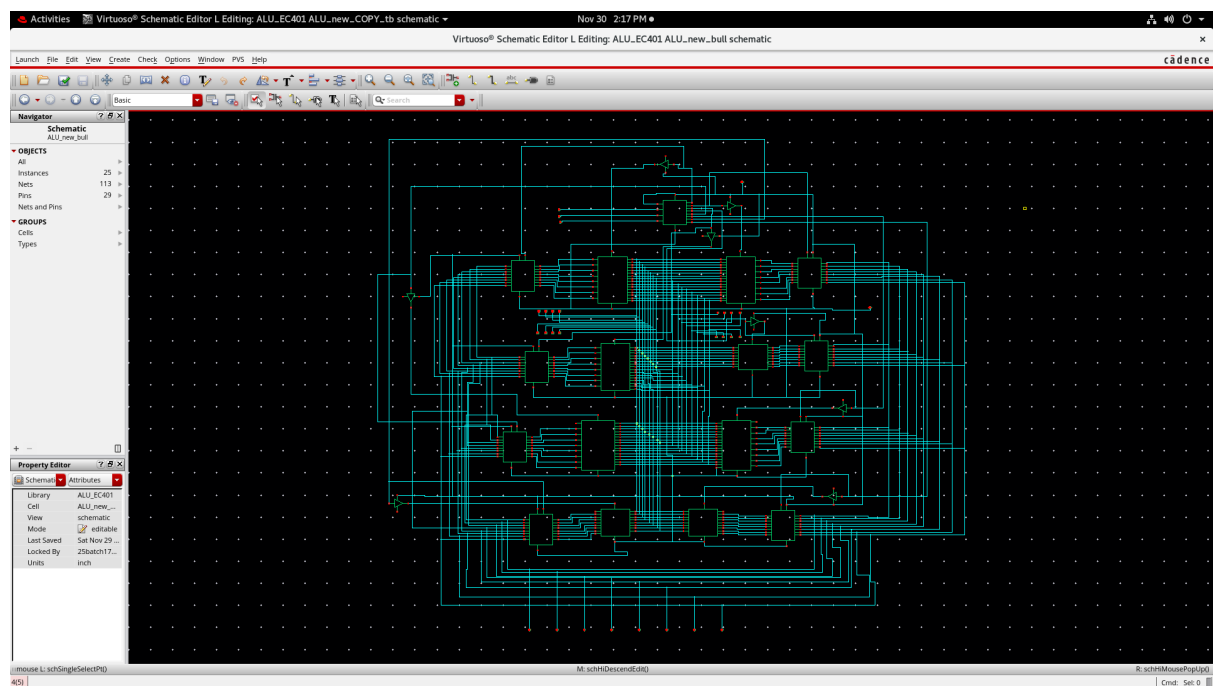


Figure 7.3: Complete DEMUX-Based Power-Gated ALU

7.5 Advantages

Advantages include:

- Reduced switching activity
- Reduced dynamic power
- Reduced propagation delay
- Elimination of multiplexer overhead
- Improved PDP

Chapter 8

Simulation Methodology

8.1 Simulation Environment

All circuits were designed and simulated using Cadence Virtuoso.

Technology	CMOS
Supply Voltage	1.8V
Simulation Type	Transient
Bit Width	8-bit
Tool Used	Cadence Virtuoso

Table 8.1: Simulation Parameters

8.2 Waveform Verification

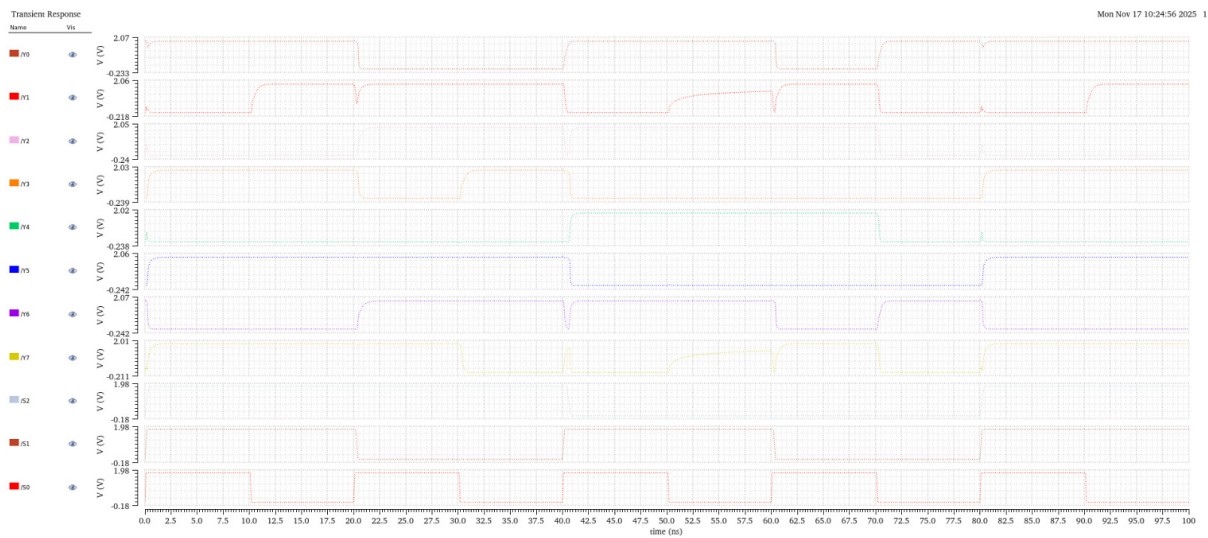


Figure 8.1: MUX-Based ALU Waveform



Figure 8.2: DEMUX-Based ALU Waveform

Chapter 9

Power Analysis

9.1 MUX-Based ALU

Average power was calculated using current integration.

$$P_{avg} = \frac{\int_0^{80ns} I(t) \times 1.8dt}{80ns}$$

Measured results:

- Average Power = 0.1707mW
- Propagation Delay = 2.614ns
- PDP = 0.4462pWs

Measured Avg Power draw of MUX based design:

$$\text{For } 80\text{ns}$$

$$P_{avg} = \int_0^{80\text{ns}} \frac{I(t) \times 1.8}{80\text{ns}} dt$$

$$\Rightarrow P_{avg} = \frac{13.66 \times 10^{-12}}{80 \times 10^{-9}}$$

$$\Rightarrow \boxed{P_{avg} = 0.1707 \text{ mW}}$$

Measured $t_p = 2.614\text{ns}$ (linear addition of individual propagation delays)

PDP for MUX based design

PDP = 0.4462pWs

Figure 9.1: MUX-Based Power Calculation

9.2 DEMUX-Based ALU Power Analysis

For accurate estimation of average power consumption of the proposed ALU, the power contributions of individual subsystems were measured separately.

The total power consumption was divided into:

- Power consumed by DEMUX power-gating circuitry
- Power consumed by buffers
- Power consumed by transmission-gate based TRUTH blocks

9.2.1 Power Draw by DEMUX

The current through the VDD terminal of the DEMUX was measured.

$$P_{avg} = \frac{\int_0^{80\text{ns}} I_{demux}(t) \times 1.8 dt}{80\text{ns}}$$

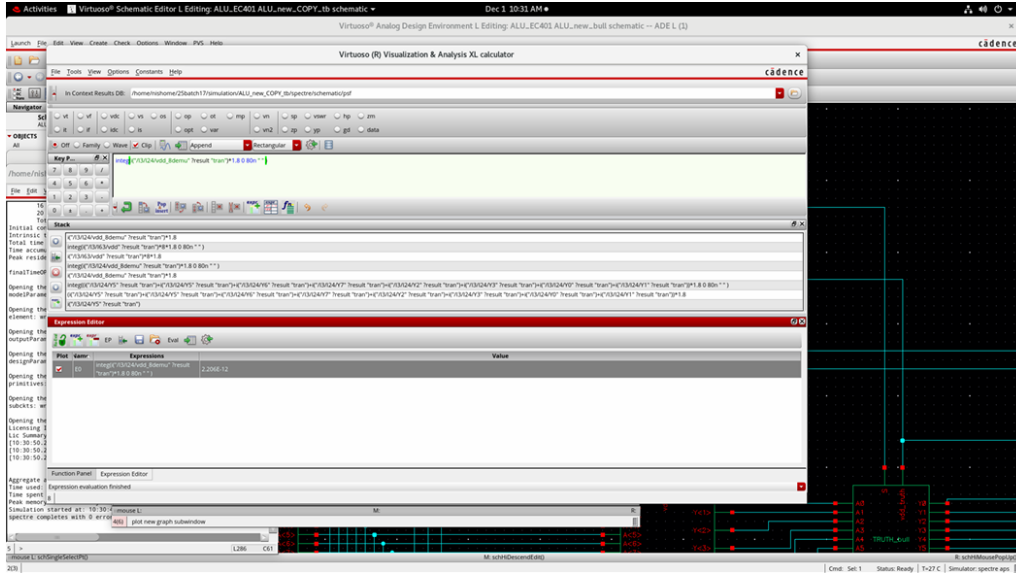


Figure 9.2: DEMUX Power Measurement Setup

$$\begin{aligned}
 &\text{Avg Power through demux} \\
 &= \frac{\int_0^{80n} I_{\text{demux}}(t) \times 1.8 \, d(t)}{80n} \\
 &P_{\text{avg demux}} = \frac{2.206 \times 10^{-12}}{80 \times 10^{-9}}
 \end{aligned}$$

Figure 9.3: Average DEMUX Power Calculation

9.2.2 Power Consumed by Buffers

The worst-case buffer power was estimated by measuring static current caused due to mismatch between input and output voltages.

$$P_{\text{avg(buffer)}} = 8 \times \frac{\int_0^{80ns} I_{\text{buffer}}(t) \times 1.8 dt}{80ns}$$

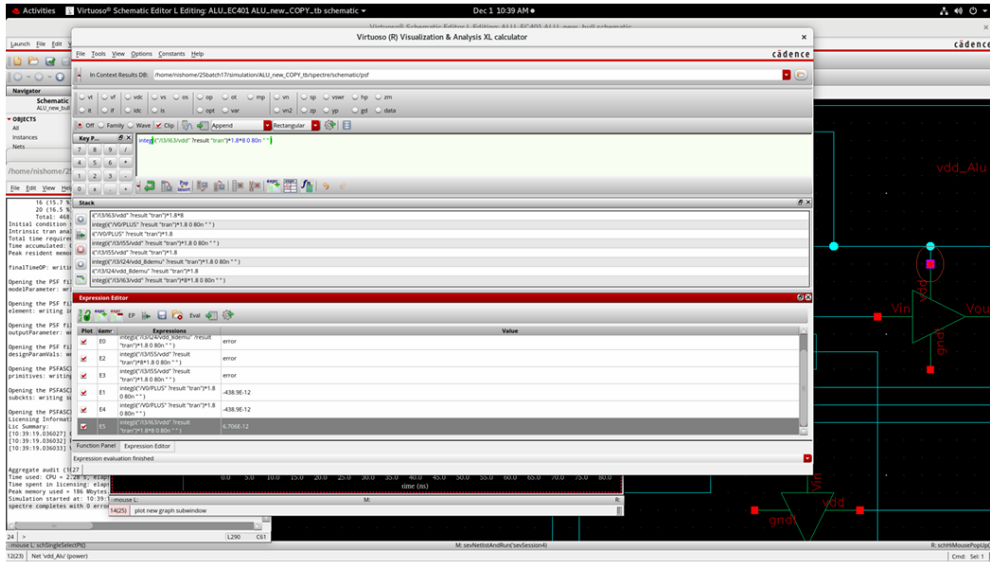


Figure 9.4: Buffer Power Measurement Testbench

Avg Power of worst case buffer:

$$P_{avg} = \frac{8 \times \int_0^{80n} I_{buffer}(t) \times 1.8 dt}{80n}$$

$$P_{avg} = \frac{6.706 \times 10^{-12}}{80 \times 10^{-9}}$$

Figure 9.5: Buffer Power Calculation

9.2.3 Power Consumed by TRUTH Blocks

The average power consumed by transmission-gate based TRUTH blocks was measured separately.

$$P_{avg(TRUTH)} = 8 \times \frac{\int_0^{80ns} I_{TRUTH}(t) \times 1.8 dt}{80ns}$$

$$P_{avg_TRUTH} = \frac{\int_0^{80n} I_{(TRUTH)}(t) \times 1.8 \, dt}{80 \times 10^{-9}} \times 8$$

$$P_{avg_TRUTH} = \frac{1.448 \times 10^{-12}}{80 \times 10^{-9}}$$

Figure 9.6: TRUTH Block Power Calculation

9.2.4 Final Average Power

The total average power was calculated as:

$$P_{final} = P_{demux} + P_{buffer} + P_{TRUTH}$$

$$P_{final} = 0.1294mW$$

Measured propagation delay:

$$t_p = 2.2884ns$$

Power Delay Product:

$$PDP = 0.1294mW \times 2.2884ns$$

$$PDP = 0.2961pWs$$

$$P_{avg\ final} = P_{avg\ (demux)} + P_{avg\ (TRUTH)} + P_{avg\ (buffers)}$$

$$\Rightarrow P_{avg\ final} = \frac{(6.702 + 2.206 + 1.448) \times 10^{-12}}{80 \times 10^{-9}}$$

$$P_{avg\ final} = 0.1294\ mW$$

Figure 9.7: Final Average Power Calculation

Chapter 10

Results and Comparative Analysis

10.1 Comparative Analysis

Table 10.1: Comparison Between Traditional and DEMUX-Based ALU

Parameter	MUX-Based ALU	DEMUX-Based ALU
Average Power	0.1707mW	0.1294mW
Propagation Delay	2.614ns	2.2884ns
PDP	0.4462pWs	0.2961pWs
Switching Activity	High	Reduced
Multiplexer Usage	Required	Eliminated
Power Gating	No	Yes

10.2 Observed Improvements

The proposed DEMUX-based ALU demonstrated:

- Reduced switching activity
- Lower dynamic power consumption
- Reduced propagation delay
- Significant improvement in Power Delay Product

Observed reduction in PDP:

33.6%

Chapter 11

Advantages, Limitations and Applications

11.1 Advantages

- Lower dynamic power consumption
- Reduced switching activity
- Improved PDP
- Elimination of multiplexers
- Reduced hardware overhead

11.2 Limitations

- Sensitivity to floating nodes
- Possible short distortions due to noise
- Output instability during capacitor discharge

11.3 Applications

- Low-power processors
- Embedded systems
- Battery-powered electronics
- Energy-efficient VLSI systems

Chapter 12

Conclusion and Future Work

12.1 Conclusion

This project successfully demonstrated the design and analysis of a DEMUX-based power-gated ALU architecture.

The proposed architecture selectively powers only the required arithmetic or logical block, thereby reducing switching activity and eliminating multiplexer overhead.

Simulation results confirmed:

- Reduced power consumption
- Reduced propagation delay
- Improved Power Delay Product
- Reduced switching activity

The proposed design achieved approximately:

33.6%

improvement in PDP compared to the traditional multiplexer-based implementation.

12.2 Future Work

Future improvements may include:

- Layout implementation
- Post-layout verification
- Glitch optimization

- Higher-bit ALU extensions
- Integration into processor datapaths

Bibliography

- [1] Neil H. E. Weste and David Harris, *CMOS VLSI Design*, Pearson Education.
- [2] M. Morris Mano, *Digital Design*, Pearson Education.
- [3] Jan M. Rabaey, *Digital Integrated Circuits*, Prentice Hall.
- [4] Sung-Mo Kang and Yusuf Leblebici, *CMOS Digital Integrated Circuits*, McGraw-Hill.
- [5] Sedra and Smith, *Microelectronic Circuits*, Oxford University Press.